UnmixDiff: Unmixing-Based Diffusion Model for Hyperspectral Image Synthesis

Yang Yu, Erting Pan, Yong Ma, Xiaoguang Mei, *Member, IEEE*, Qihai Chen, and Jiayi Ma, *Senior Member, IEEE*

***Abstract*— The scarcity of hyperspectral images (HSIs) hinders the development of processing methods and downstream applications. HSI synthesis, which aims to generate realistic samples from the existing datasets, is undoubtedly a prospective and economical solution for the HSI data shortage problem. Inspired by the impressive performance of the diffusion model (DM) in image synthesis tasks, this article initiatively proposes an unmixing diffusion (UnmixDiff) model for high-quality HSI generation. The method starts with training an unmixing network to learn the distribution characteristic of objects (abundance). By incorporating the unmixing autoencoder into the DM, the UnmixDiff transforms the HSI generation into the abundance domain, which maintains the consistency of the generated spectral profile, reduces the computational complexity, and introduces a clear physical interpretation into the hyperspectral image synthesis tasks. After that, we construct a diffusion generation model in abundance space to generate realistic abundance maps. Instead of synthesizing original hyperspectral images, the proposed UnmixDiff synthesizes abundance maps to simulate objects' distribution rather than superficial textures. In this way, realistic HSI samples are generated by mixing the synthesized abundance with the scene end-members. With the comparative experiments, the proposed method achieves state-of-the-art performance in HSI synthesis tasks, effectively alleviating the HSI data scarcity and supporting widespread HSI applications. The code is available at https://github.com/yuyang95/UnmixingDM.**

***Index Terms*— Diffusion model (DM), hyperspectral DM, hyperspectral image (HSI) synthesis, hyperspectral unmixing (HU).**

I. Introduction

HYPERSPECTRAL images (HSIs) have broad applications for their high spectral resolution, ranging from large-scale Earth observation to micromedical imaging [1], [2], [3]. With the increasing use of data-driven deep learning (DL) methods, there is a surge in demand for high-quality and diverse HSIs [4]. Although the extensive applications of HSIs have driven the advancement of HSI sensors and hyperspectral data availability like PRISMA [5] and EnMAP [6], [7], as well as various snapshot hyperspectral imaging devices [8], [9], building large-scale HSI datasets with desired characteristics, diversity, and high quality would be costly and time-consuming. Fortunately, compared to collecting data using HSI sensors, image synthesis provides an economical alternative to generate realistic images, becoming a hotspot research field with diverse applications, e.g., training samples supplement and aided design [10], [11]. Typically, the existing HSI synthesis methods mainly include three categories: physical modeling-based methods [12], [13], [14], [15], affine transformation-based methods [16], and spectral reconstruction-based methods [17], [18], [19]. However, the current methods have apparent disadvantages.

Specifically, physical modeling-based methods, which aim to generate HSIs by simulating the propagation of light, material properties, and corresponding physical processes, could utilize satellite images to derive bio/geophysical parameters that are then ingested into a physical model to generate HSIs with realistic spatial patterns. However, due to the real-world's complexity, these methods often require sophisticated design and offer relatively weak generalization and feasibility. The affine transformation-based methods, such as rotation, flipping, and scaling, serve as prevalent tools for data augmentation in DL. Despite these operations being easy to implement, these approaches are limited in their capacity to enhance the diversity of the dataset. The spectral reconstruction-based methods aim to increase the spectral resolution of RGB or multispectral (MSI) images. The training of the spectral reconstruction network usually requires paired RGB-HSI or MSI-HSI images [18]. In addition, such methods only increase the information of existing data in the spectral dimension and do not generate new data. In particular, some surface materials have similar or identical colors with completely different spectral characteristics. Therefore, the obtained HSIs through RGB spectral reconstruction cannot guarantee the reliability of the spectral profile. To sum-up, the limitations of the existing HSI synthesis methods lie in the reliability and diversity of the synthesis HSIs [20].

Inspired by the excellent performance of the deep generative model in RGB image synthesis [21], [22], a natural way is introducing the deep generative model into HSI synthesis to overcome the limitations of the existing HSI synthesis methods. Deep generative models, such as variational autoencoders (VAEs) [23], [24] and generative adversarial networks (GANs) [25], [26], aim to simulate the distribution of training samples [27]. Recently, diffusion models (DMs) [28], [29]

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have shown impressive performance in high-quality image generation. As an emerging generative model, the DMs overcome the drawbacks of other methods. For example, VAEs often yield suboptimal sample quality [30], and GAN-based methods need sophisticated regularization and training [31]. In general, the DM can generate realistic RGB samples with rich diversity. Therefore, it is a promising manner to introduce the DM into the HSI synthesis tasks.

Although the DMs have shown desirable performance in RGB image synthesis, it is challenging when applied to the HSI generation. Compared to RGB image synthesis, generating high-quality HSI with realistic spatial and spectral features is more difficult due to its high spectral dimension and the unreliability of the synthesis spectral profile. Specifically, the DMs estimate the noise level based on the U-Net [32] architecture to preserve the dimensional consistency [33]. However, the high dimensions of HSI increase the computation complexity and the difficulty of noise estimation, which result in the poor performance in spatial details. In addition, there are huge differences between the synthesis and actual spectral profiles. Therefore, the spectral profile synthesized directly by DMs may not be physically significant and represent the spectral characteristics of any surface materials. Moreover, we cannot perceive the surface materials of generated objects in the synthetic images.

To solve the aforementioned problems, we develop an unmixing diffusion (UnmixDiff) model, which tackles the high-dimensional HSI synthesis problem by incorporating the unmixing autoencoder and the diffusion generation model. Specifically, the overall framework consists of two parts: the unmixing autoencoder, which decomposes the original HSI into a combination of the intrinsic object distributions (abundance maps) and corresponding spectral features (end-members) [34], and the diffusion process, which aims to simulate the distribution of abundances obtained from the previous unmixing process. By transforming the diffusion process from the image domain into the abundance domain, the synthesis objective of the DM has shifted from high-dimensional hyperspectral images to low-dimensional abundance maps.

Compared to the basic DM, the proposed UnmixDiff has many advantages. First, by introducing the unmixing process, we can build the diffusion process on abundance maps, whose dimensions are far less than the original HSI. It greatly decreases the computation complexity and increases the precision of noise inferences at the same time. Second, the synthetic abundance maps simulated the mixed pixel phenomenon under different imaging situations with shared end-members, which avoids generating a meaningless spectral profile. Moreover, the unmixing process perceives the surface materials in an unsupervised manner. Third, the unmixing process can obtain the actual spectral reflectance of surface materials (end-members) from the original HSI. During the synthetic stage, we synthesize the HSIs by combining the synthetic abundances with the actual spectral reflectance rather than synthesizing the variant spectral profile influenced by the environment. The overall framework can eliminate the spectral variability problem, which exists extensively in HSIs [35].

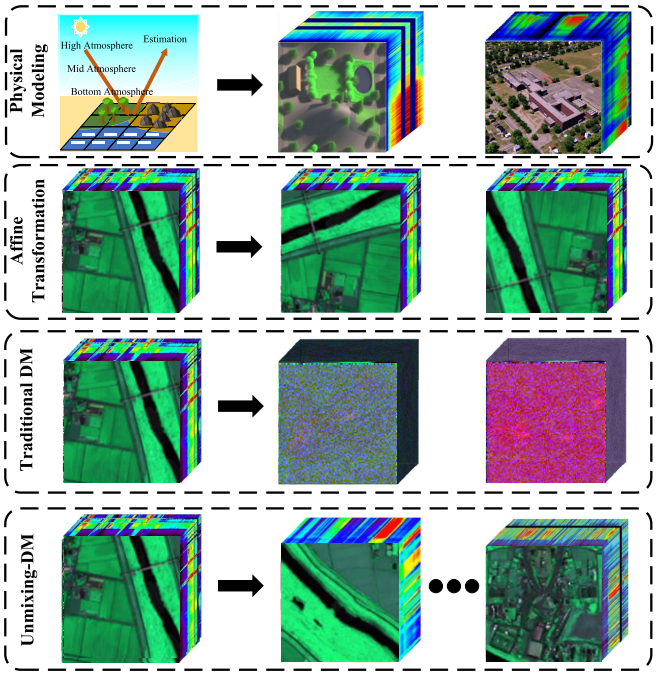


Fig. 1. Comparison of the existing HSI synthesis methods. Physical-based HSI lacks realistic data, and the transformation-based methods do not increase diversity. The proposed UnmixDiff can generate new realistic data with rich diversity.

Compared to the existing HSI synthesis methods, the proposed UnmixDiff can generate high-quality HSIs with realistic spatial details, reliable spectral features, and rich diversity. A brief comparison of the existing HSI synthesis methods is shown in Fig. 1. Moreover, the proposed UnmixDiff overcomes the feasibility issues of the deep generative model in the hyperspectral image synthesis field. The contributions are as follows.

1) We conduct a robust and generalized unmixing model and integrate it into the synthetic process. The proposed unmixing model achieves better generalization, unmixing accuracy, and a higher compatibility with downstream tasks. The integration of the unmixing model not only reduces the complexity of generating hyperspectral images but also ensures the spectral consistency and physical meaning.
2) We develop an abundance-based DM for hyperspectral image synthesis, capable of producing high-quality HSIs. The proposed model effectively addresses the limitations of the current diffusion-based generative models in HSI synthesis tasks. The UnmixDiff can generate HSIs with realistic spatial details while maintaining the spectral consistency. Moreover, the proposed model is not limited to hyperspectral image generation but can be extended to other DM-based tasks [36].
3) We conduct the downstream expansion experiments based on the synthetic HSI dataset, demonstrating the beneficial impact of synthetic data on downstream tasks. Various experiments demonstrate the feasibility and reliability of the proposed UnmixDiff in HSI synthesis tasks.

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II. Related Works

A. Deep Generative Model-Based RGB Image Synthesis

Over the past few years, image synthesis has achieved impressive progress owing to the development of the deep generative model [37], [38]. The GAN has shown the good perceptual quality of the synthetic image by adversarial learning [39], [40]. However, the GAN is hard to train because of the difficulty in finding the balance of the generator and discriminator. In undesirable training situations, the GAN shows instability in synthetic quality. Compared to the GAN, the VAE, a likelihood-based method, renders optimization performance with good distribution estimation. However, the VAE-based generative model often yields suboptimal generation performance.

Recently, the DM has achieved state-of-the-art results in image synthesis tasks [33], [41]. Compared to other generation models, the DM offers desirable properties such as a clear training objective, model stability, and easy extensibility. Although the DM shows preferred performance in image generation, the training of DM is a slow and computationally expensive process. Some methods develop the two-stage synthesis framework by combining the compression and the generative models [23], [42]. The two-stage approaches decrease the computational complexity by projecting the raw image into latent space [30], [43]. For those approaches, the synthesis quality is influenced by the performance of the generative model and the compression model.

B. Hyperspectral Image Synthesis

Although deep generative models have demonstrated remarkable performance in synthesizing RGB images, the existing deep generative model-based HSI processing has mainly focused on other tasks, e.g., HSI classification [44], [45], denoising [46], super resolution [47], and the like [48], [49]. The deep generative model-based new hyperspectral image synthesis lacks substantial progress or achievements [50]. Although some researchers have attempted to introduce the GAN for synthesizing spectral profiles without spatial information [19], synthesizing high-quality new HSIs with realistic spatial–spectral features is still a challenging task. HSI synthesis is severely challenged due to high spectral dimensionality, and simply migrating the generative model for new HSI synthesis could result in meaningless results.

The existing HSI synthetic methods generate HSIs in different ways, including physical modeling, affine transformation, and spectral reconstruction [16], [51]. Physical modeling-based methods, such as empirical and statistical approaches, try to synthesize HSIs based on the simulation of physical phenomenon [52]. Typically, collecting the radiometric data through surrogate data and combining the information with satellite input for synthesizing the desired HSIs is a commonly used method and exhibits meaningful applications [13], [14], [15]. However, collecting paired multisource HSIs is time-consuming and expensive [20]. Statistical approaches use satellite images to derive the geophysical parameters [53], [54] and then integrate them into the physical model (such as Monte Carlo models) to generate HSI with realistic spatial patterns [55], [56]. Furthermore, the statistical algorithms can be used for 3-D scene simulation based on the LiDAR data [57]. However, such approaches are very complex to parameterize, and the accuracy of the key parameters highly influences the synthetic performance. Moreover, the inherent complexity of the natural world dramatically affects the generalizability, feasibility, and authenticity of the physical model-based methods [20]. The affine transformation, which cannot expand the diversity, is extensively used for data augmentation. Some methods take the RGB image as conditional constraints to generate HSI [18]. Those methods essentially are the spectral resolution or RGB–HSI fusion tasks, without generating new scenes. Moreover, although the mentioned approaches can be roughly regarded as synthetic methods for addressing data scarcity problem, there is a significant difference between these methods and the deep generative model-based synthesis. For the physical modeling-based methods, those methods need collecting the radiation characteristics of a specific area, using the root mean square error (RMSE) to evaluate the synthesizing performance. For the transformation-based methods, those methods do not evaluate the augmentation performance. The spectral reconstruction-based methods are utilized for expanding the spectral information under supervised training with ground truth. Reconstruction metrics are used to assess reconstruction quality. Instead, the deep generative model-based synthesis methods are appraised by the reliability and diversity of the synthetic HSI samples. Compared to the constantly emerging methods in RGB image synthesis, only few methods focus on deep generative model-based hyperspectral image synthesis [58]. Moreover, the existing deep generative-based HSI synthetic methods have obvious limitations regarding the synthetic spectral reliability and the wavelength range.

C. Hyperspectral Unmixing

Recently, DL has emerged as a powerful and effective tool for hyperspectral unmixing (HU) due to its capability to learn the complex relationships between the observed spectral data and the underlying materials [59], [60]. DL-based unmixing methods, which often utilize autoencoders as the backbone, have achieved superior performance in solving the HU problem [61], surpassing the conventional geometric [62], statistical [63], and non-negative matrix factorization-based unmixing approaches [64]. Various researchers have designed effective network architectures based on the unmixing autoencoder to enhance the unmixing performance. The basic autoencoder consists of two parts: the encoder, which aims to learn the low-dimensional representation (abundance), and the decoder, which reconstructs the HSI with the corresponding basis (end-members) [65]. Considering the lack of spatial information in basic autoencoders, CNN-based unmixing autoencoders have been proposed to improve the spatial presentation of the HU network [66]. Specifically, adversarial learning has been introduced to optimize the network weights by learning the statistical properties of input HSI [34]. Those approaches have achieved desirable performance in unmixing areas [67].

In this article, the unmixing process differs from traditional HU tasks. From the perspective of task definition, the existing unmixing methods can be regarded as an independent end-to-end task, aiming to perceive the distribution of materials

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and their compositions [59]. In the HSI synthesis tasks, the unmixing process is considered to be part of the image generation tasks. The unmixing network not only encompasses the objectives of traditional unmixing tasks but also serves as a dimensionality reduction method with physical significance to meet the requirements of the abundance DM. From the perspective of training style, the prevailing unmixing approaches employ single HSI for training. The network training is conducted with the reconstruction error of a single image as a constraint, and different parameters are set for different images. Therefore, the existing unmixing methods typically have a relatively simple network design [60], [67]. In contrast, in the synthesis tasks, the unmixing training incorporates many HSIs as the training data to derive a set of parameters that are universally applicable across the entire dataset. This training style fosters a more robust and adaptable model capable of handling a diverse range of HSIs. Due to different task objectives and training styles, the existing unmixing methods cannot meet the requirements when migrating the current unmixing models to the synthesis task.

III. Methodology

A. Motivation

Due to the high spectral dimensionality and unique physical meaning of the spectral profile, generating high-quality hyperspectral imagery (HSI) is extremely challenging. Therefore, we introduce the unmixing process into the DM to project the synthesis tasks from HSI space to abundance space. Based on the linear mixing model (LMM) [34], the HSI is represented by a linear combination of abundance maps and end-members. It can be formulated by the following equation:

$$\mathbf{Y} = \mathbf{AE} \tag{1}$$

where $\mathbf{Y} \in \mathbf{R}^{HW \times B}$ is the HSI with H height, W width, and B bands. $\mathbf{A} \in \mathbf{R}^{HW \times p}$ indicates the abundance matrix and $\mathbf{E} \in \mathbf{R}^{p \times B}$ is the end-member matrix with p end-members. By introducing the unmixing process, the synthetic targets will be transformed into synthesizing abundance maps, whose dimensions are less than the original HSIs in most cases. The synthetic targets can be formulated as follows:

$$\tilde{\mathbf{A}} = \mathcal{G}(z), \quad \tilde{\mathbf{A}} \sim \mathbf{A} \tag{2}$$

where $\mathcal{G}$ is the trained DM and z is the input random noise. $\tilde{\mathbf{A}} \sim \mathbf{A}$ represents that they have a similar distribution. Therefore, we can obtain the synthetic HSI by combining the synthesized abundance maps $\tilde{\mathbf{A}}$ and end-members $\mathbf{E}$. It can be formulated as follows:

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{A}}\mathbf{E} \tag{3}$$

where $\tilde{\mathbf{Y}}$ represents the synthetic HSIs.

B. Overall Framework

Based on the mentioned above, the proposed UnmixDiff contains two parts: the unmixing autoencoder and the abundance DM. The training process starts with training an unmixing autoencoder to obtain the low-dimensional abundance maps and perceive the end-member spectrum. Introducing the unmixing autoencoder into the DM solves the dimensionality curse and ensures the synthetic HSIs' reliability. After that, the training of the diffusion generation model is performed on the abundance maps to learn the context relationship, distribution characteristic, and mixing manner of objects. In the synthesis process, the synthetic abundance maps generated by the DM will be put into the decoder part of the unmixing network. The synthetic HSI can be regarded as the simulation of surface materials under different spatial distribution and mixing levels. Subsequently, the proposed UnmixDiff can generate high-quality HSIs with rich diversity. The overall framework is shown in Fig. 2.

C. Unmixing Autoencoder

Given the substantial differences in task objectives, training strategies, and model design between the unmixing process in the HSI synthesis framework and traditional unmixing tasks (as mentioned in Section II-C), the existing unmixing network cannot meet the synthetic requirements. Unlike the existing unmixing networks which are trained on a single HSI, our network is trained on several HSIs to learn a general network weight. This training strategy guides the entire network design and greatly enhances the generalization performance of the unmixing network. At the same time, it ensures that the unmixing network can be well integrated into the generative network and guide it to generate meaningful HSIs. In this article, we construct the unmixing autoencoder architecture based on the residual spectral attention (RSA) module. The overall unmixing framework is shown in Fig. 3.

The encoder is composed of several RSA modules and a softmax layer, aiming for feature extraction and dimensional reduction. The RSA contains some basic components, including the convolution layer, the GeLU [68] activate function, the channel attention layer, and the layer normalization. The arrangement details are shown in Fig. 4.

Considering the high spectral dimension and band redundancy of HSI, the channel attention layer aims to capture the channel-wised relationship [69]. Layer normalization is introduced to stabilize the input distribution and accelerate convergence. The encoder ends up with a softmax layer, which aims to ensure the abundance sum-to-one constraint (ASC) and the abundance non-negative constraint (ANC) [67]. The abundance maps are obtained from the encoder output, whose dimensions equal the number of end-members [65]. The decoder contains a 1×1 convolutional layer, which is designed to combine the end-members and abundances linearly. The 1×1 convolutional decoder layer simulates the LMM, of which the weights represent the extracted end-members [66].

The loss function of the proposed unmixing framework comprises three parts: the pixel-wise reconstruction error, the reconstruction spectral angle distance (rSAD), and the end-member total variation (TV).

The pixel-wise L_1 distance is introduced to guarantee the similarity of the inputs and reconstructed HSIs. It can be formulated by the following equation:

$$\text{Loss}_{L1} = \left\| Y - \widehat{Y} \right\|_1. \tag{4}$$

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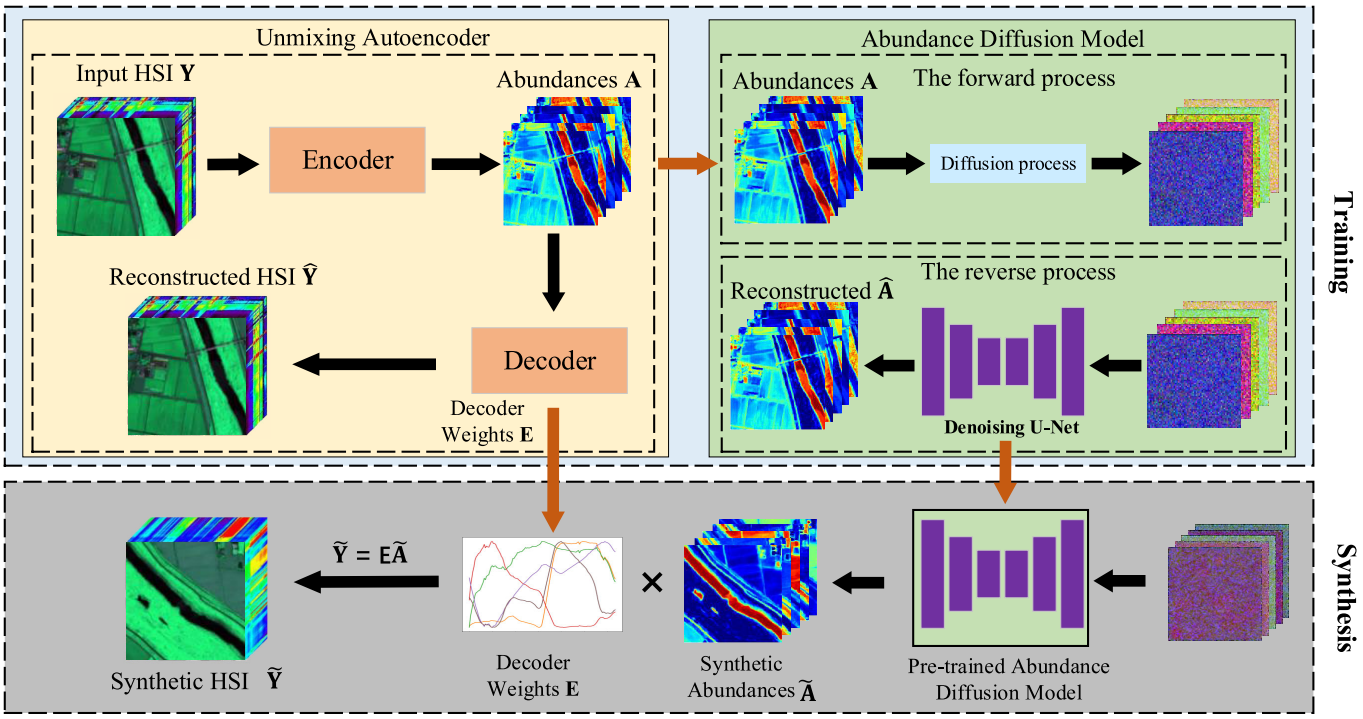


Fig. 2. Overall framework of the proposed UnmixDiff. In the training phase, the abundances obtained from the unmixing autoencoder will apply to the DM training. In the generation phase, the synthetic abundances generated by the pretrained DM will combine with decoder weights (end-members) to generate HSIs. The orange arrow represents the information sharing.

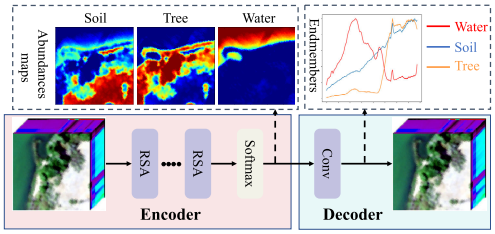


Fig. 3. Unmixing framework. The abundances arise as the encoder result, and the end-members the decoder weight.

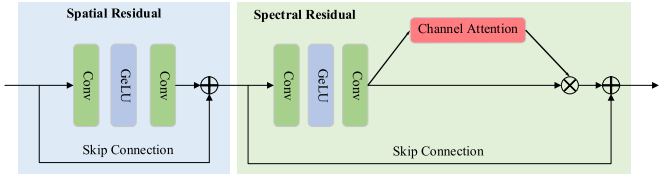


Fig. 4. Framework of the RSA module.

The SAD aims to maintain the spectral feature of the reconstructed HSI. It can be formulated by the following equation:

$$\text{Loss}_{\text{SAD}} = \arccos\left(\frac{\langle Y, \widehat{Y}\rangle}{\|Y\|_2\|\widehat{Y}\|_2}\right). \tag{5}$$

When training an unmixing network, traditional unmixing networks often use end-member extraction methods (such as VCA [62]) to obtain the initial end-members of the HSI and use the extracted end-members to initialize the decoder weights. However, using end-member extraction methods to initialize network weights requires additional processing, which not only makes network training nonend-to-end but also causes error accumulation. Moreover, due to the training requirement of the proposed unmixing framework, the traditional end-members initialization methods are not applicable in our network. In this situation, the decoder weights, that is, the end-members, exhibit spectral spikes because of the inadequacy of initialization. Therefore, we introduce the spectral TV loss to preserve the spectral smoothness of the extracted end-members. It can be formulated by the following equation:

$$\text{Loss}_{\text{TV}} = \sum_{i}^{B}(e_{i+1} - e_i) \tag{6}$$

where e_i is the end-members' reflectance on the ith band, and B represents the number of bands. In summary, the total loss function of the proposed unmixing is a weighted sum of the three parts

$$\text{Loss} = \text{Loss}_{L1} + \alpha\text{Loss}_{\text{SAD}} + \beta\text{Loss}_{\text{TV}} \tag{7}$$

where α and β are used to balance the contributions of different terms. We set $\alpha = 0.1$ and $\beta = 0.01$ in this article.

D. Abundance DM

The DM contains two parts: the forward process, named diffusion process, and the reverse process. Both forward and reverse processes can be defined as a parameterized Markov chain, and the reverse process can be used for sample generation.

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1) Forward Diffusion Process: We define the abundance maps before adding noise as A_0, $q(A_0)$ is the distribution of A_0, and the abundance maps after t times of adding Gaussian noise as A_t. The forward diffusion process refers to gradually adding Gaussian noise to the abundance maps in T steps, producing a sequence of noisy abundance maps $A_0, \ldots, A_T$. The standard deviation of the noise is determined by the fixed value β_t, and the expectation is determined by the fixed value β_t and the data A_t. This process can be regarded as a Markov chain process, where the distribution at time t depends on the distribution at time $t-1$. With the increasing of t, the data A_T eventually becomes an anisotropic Gaussian distribution. According to the property of Markov chain, the distribution $q(A_T)$ can be formulated as follows:

$$q(\mathrm{A}_t \mid \mathrm{A}_{t-1}) = \mathcal{N}\left(\mathrm{A}_t; \sqrt{1-\beta_t}\,\mathrm{A}_{t-1}, \beta_t\,\mathrm{I}\right) \tag{8}$$

$$q(\mathrm{A}_{1:T} \mid A_0) = \prod_{t=1}^{T} q(\mathrm{A}_t \mid A_{t-1}). \tag{9}$$

According to the reparameterization [70], the value of A_t at any given time step t can be expressed as a function of A_0 and β_t. We define that $\alpha_t = 1-\beta_t$ and $\overline{\alpha_t} = \prod_{i=1}^{t} \alpha_i$, and $q(A_t \mid A_0)$ can be formulated as follows:

$$A_t = \sqrt{\bar{\alpha}_t} A_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \tag{10}$$

$$q(A_t \mid A_0) = \mathcal{N}\left(A_t; \sqrt{\overline{\alpha_t}} A_0, (1-\overline{\alpha_t})\mathbf{I}\right) \tag{11}$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.

2) Reverse Process: The reverse process aims to reconstruct the abundance maps A_0 from Gaussian noise A_T and conditional constraint x. We can assume that it is also a Gaussian distribution, which cannot be fitted step by step [28]. Therefore, a parameter distribution needs to be constructed for estimation. If we can get the distribution of the reverse process $q(A_{t-1} \mid A_t)$, we can generate a real sample by gradual denoising from the random noise $A_{\mathbf{T}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ [71]. Since $q(A_{t-1} \mid A_t)$ depends on the entire data distribution and cannot be directly calculated, we use a DL-based model to approximate the reverse distribution p_θ. According to the property of Markov chain, (8), and (9), the reverse distribution p_θ can be formulated as follows:

$$p_\theta(A_{0:T}|x) = p(A_T) \prod_{t=1}^{T} p_\theta(A_{t-1}|A_t, x) \tag{12}$$

$$p_\theta(A_{t-1}|A_t, x) = \mathcal{N}(A_{t-1}; \mu_\theta(A_t, x, t), \Sigma_\theta(A_t, x, t)). \tag{13}$$

Based on the Bayes rules, we can calculate the posterior $q(A_{t-1} \mid A_t, A_0)$ by β_t and $\tilde{\mu}_t(A_t, A_0)$, which can be defined as follows:

$$\tilde{\beta}_t = 1/\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\overline{\alpha}_{t-1}}\right) = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t} \cdot \beta_t \tag{14}$$

$$\tilde{\mu}_t(A_t, A_0) = \frac{\sqrt{\alpha_t}(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t} A_t + \frac{\sqrt{\overline{\alpha}_{t-1}}\beta_t}{1-\overline{\alpha}_t} A_0. \tag{15}$$

According to (10), A_t can be expressed as $A_t(A_0, \epsilon) = \sqrt{\bar{\alpha}_t} A_0 + (1-\bar{\alpha}_t)^{1/2}\epsilon$ for $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. In this situation, we can design a DL network to predict the noise ϵ with inputs of A_t and t. We obtain the expectation as follows:

$$\mu_\theta(A_t, x, t) = \frac{1}{\sqrt{\alpha_t}}\left(A_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(A_t, x, t)\right). \tag{16}$$

Finally, we can obtain accurate noise estimation by constructing (17) as the training objective, and the detailed solutions and derivation process can be referred to [28]

$$L_{t-1} = \mathbb{E}_{A_0, x, \epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t} A_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, x, t\right)\right\|^2\right]. \tag{17}$$

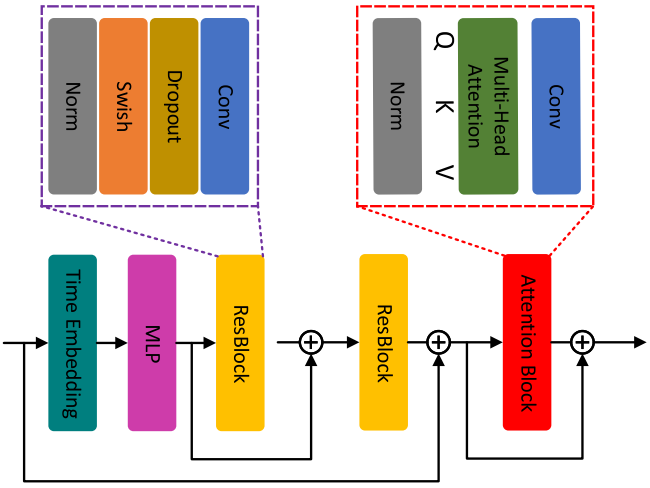


Fig. 5. Framework of residual attention module.

The reverse process aims to remove the noise sequentially estimated by a neural network from the random noise. Similar to the existing DM-based methods [28], [33], [41], we chose the U-Net structure as the basic backbone and the residual attention network as the feature extraction module to conduct the denoising network. The denoising architecture consists of a sequential arrangement of residual attention modules and downsampling layers in the encoder part, followed by a stack of residual attention modules with upsampling layers in the decoder. Skip connections are employed to establish direct connections between layers of the same spatial size. The framework of the residual attention module is shown in Fig. 5. The residual attention module begins with a time embedding operation [72], which aims to encode the time steps into the network. Then, an MLP module is chosen to conduct the nonlinear transformations on the features, enhancing the overall model expression capability. Next are two residual modules, each consisting of a normalization layer, a swish activation function [73], a dropout layer, and a convolution layer. Finally, the self-attention mechanism, which consists of normalization, multihead attention, and a convolutional layer, is introduced to capture the long-range dependency and global information [72]. Additionally, the entire module incorporates multiple residual connections.

After completing the training of the abundance-based DM, we can generate high-quality abundance maps using the pre-trained DM. The synthesis process takes random noise as input, with the same dimensionality as the training abundance

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Algorithm 1 Training for Abundance-Based Diffusion

Input: The abundance maps $A_0 \in \mathcal{R}^{H\times W\times N}$, condition x

// N is the number of end-members.

while *not converged* **do**

  $(A_0, x) \sim p(A, x)$;

  $t \sim \text{Uniform}(\{1, \ldots, T\})$;

  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$;

  Gradient descent

  $\nabla_\theta \left\| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} A_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, x, t) \right\|^2$;

end

Algorithm 2 Synthesizing Abundance Maps

Input: random noise $A_T \in \mathcal{R}^{H\times W\times N}$

Output: synthetic abundance maps $\tilde{A}_0 \in \mathcal{R}^{k\times W\times H}$

$A_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$;

for $t = 1 \quad to \quad T$ **do**

  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$;

  $\tilde{\mathbf{A}}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\tilde{\mathbf{A}}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\tilde{\mathbf{A}}_t, x, t) \right) + \sigma_t \mathbf{z}$;

end

return $\tilde{A}_0$

maps. We use the trained DM to denoise the initial noise maps, gradually approaching the distribution of real data. Repeat the denoising steps many times until the noise is almost completely removed. Consequently, we transform the initial random noise into high-quality abundance maps generated by the abundance-based DM. Taking the abundance maps into the unmixing decoder, we can obtain the final synthetic HSI.

IV. Experiments

In this section, we implement experiments to evaluate the feasibility and effectiveness of the proposed UnmixDiff from various perspective.

A. Experimental Setting

1) Datasets: To demonstrate the generalization ability of the proposed method, we conduct the experiments on different types of datasets to train the different UnmixDiff models, including airborne dataset, satellite-borne dataset, and ground dataset. For airborne hyperspectral dataset, we chose the Chikusei [74] dataset. The Chikusei dataset primarily comprises village scenes and encompasses various prominent features, such as water bodies, cultivated land, industrial areas, and residential structures. The Chikusei dataset has a spatial size of 2517 × 2335 with 128 spectral bands ranging from 363 to 1018 nm. Due to missing information on the edge, we crop the center region to obtain a subimage with 2304 × 2048 pixels and cut the subimage into 128 × 128 pixels with a 50% overlap rate. Finally, we obtain 837 HSIs with 128 × 128 pixels. We randomly selected 637 HSIs for training, 100 for validation, and 100 for testing. For the satellite-borne dataset, we choose the Whu-OHS [75], which contains 4803 satellite HSIs with 32 bands ranging from 400 to 1000 nm. For the ground hyperspectral dataset, we choose the NITRE22 spectral reconstruction dataset (ARAD) [76], which contains 1000 HSIs with a spatial size of 482 × 512 and 31 spectral bands ranging from 400 to 700 nm. The dataset contains complex and irregular types of ground objects, encompassing hundreds of material types, such as murals, faces, plants, and buildings. Therefore, it is challenging to unmix and generate images in this complex dataset. We follow the official dataset division (900 HSIs for training, 50 for validation, and 50 for testing) and cut the data into 128 × 128 pixels with a 50% overlap rate. For the unmixing experiments, we adopt the aforementioned data split strategy to evaluate the unmixing performance. For the synthesis experiments, we will use all the data available in the relevant dataset for training the UnmixDiff. The ablation experiments are conducted on the Chikusei dataset, and the synthetic experiments are conducted on the three mentioned datasets.

2) Ablation Experiments Setting: To demonstrate the effectiveness of the proposed UnmixDiff, we conduct ablation experiments from different aspects. All the ablation experiments are conducted on the Chikusei dataset. First, to verify the necessity of introducing the unmixing process into HSI synthesis, we compare the performance of the proposed unmixing autoencoder and the symmetric autoencoder in HSI synthesis tasks. In the symmetric autoencoder, the network structure of the encoder and decoder is usually mirror-symmetric, with an identical number of layers and neurons. Generally, the symmetric autoencoder has a similar function to the unmixing autoencoder in other tasks, in which the encoder maps the input data to a low-dimensional latent representation, and the decoder aims to reconstruct the low-dimensional representation to the original data. Therefore, we construct the experiments with a basic symmetric autoencoder to verify the necessity of introducing the unmixing process into HSI synthesis tasks.

Considering the multiple loss items of our unmixing network, we conduct experiments to analyze the parameter sensitivity analysis of the unmixing loss function. For the Loss_{SAD}, it aims to improve the spectral consistency between the reconstructed HSI and the original HSI. We conduct the Loss_{SAD} ablation experiments with an interval of 0.1. The weight of the Loss_{TV} is set to 0 during the Loss_{SAD} ablation experiment to avoid the impacts. For the Loss_{TV}, a large weight of TV loss often leads to over-smooth representations and the loss of details. Therefore, the weight of TV loss is generally small. We set the β starting from 0 and built ablation experiments with an interval of 0.005.

We also investigate the relationship between the number of end-members and the HSI generation performance. We conduct the ablation experiments when the numbers of end-members are 3, 5, 8, 10, 15, and 31, respectively. The number of end-members equals the dimension of abundance maps, which will influence the dimension of the abundance DM. Specifically, reducing the number of end-members can significantly reduce the computational cost and improve the

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efficiency of the abundance DM. However, a smaller number of end-members may result in an incomplete representation of the materials present in the scene. On the other hand, a larger number of end-members can adequately represent the materials in the scene, but it may generate meaningless abundance images and corresponding end-members. Moreover, increasing the number of end-members will increase the computational complexity of the abundance DM. Therefore, selecting an appropriate number of end-members is necessary, which can achieve a good reconstruction accuracy of the unmixing network and ensure that the abundance maps have clear physical meanings while maintaining the quality of the generated HSIs.

3) Unmixing Experiments Setting: In model training, the encoder in the unmixing network is responsible for projecting the hyperspectral image from the high-dimensional spectral domain to the low-dimensional abundance domain, which greatly reduces the computational complexity of the DM. In the process of generating new data by the trained model, the decoder aims to reconstruct the synthesis abundance maps to the HSIs. Therefore, the performance of the unmixing network is critical for the quality of the HSI generation. To demonstrate the rationality of the proposed unmixing autoencoder in HSI generation tasks, we choose three traditional unmixing networks for comparison: CNNAEU [66], DAEU [65], and MSNet [67]. Specifically, DAEU and CNNAEU employ the conventional autoencoder structure and the convolutional autoencoder structure, respectively. On the other hand, MSNet achieves state-of-the-art results in traditional unmixing tasks by incorporating multistage networks. This selection of networks is crucial for validating the effectiveness and efficiency of the proposed unmixing network. We evaluate the performance of unmixing networks from two aspects, including the precision of abundance and end-members, and the reliability of reconstructed HSIs. We use three commonly used metrics to evaluate the reliability of reconstructed HSIs, including the reconstruction root mean square error (rRMSE), the reconstruction SAD (rSAD), and the peak signal-to-noise ratio (PSNR). Because of the absence of the unmixing ground truth, we judge the reliability of the abundance maps and end-members visually.

4) Unconditional Synthesis Experiments Setting: In the image generation experiment, we compare with the RGB image generation models Basic DM [28] and latent DM (LDM) [42]. Specifically, the LDM learns a space that is perceptually equivalent to the image space by building an autoencoder model. Compared with the traditional DM model, LDM greatly reduces the amount of computation by constructing the learning potential space of the autoencoder model and has been widely used in the industrial fields. However, since training the LDM autoencoder model involves the perception loss constructed by the RGB image dataset, which cannot be applied to hyperspectral images, we delete the corresponding loss term. We comprehensively evaluate the quality of the generated HSIs from different aspects, including the spatial quality, the generation diversity, and the spectral consistency with the generated images. For assessing the quality and diversity of the synthetic HSIs, we employ commonly used image generation evaluation metrics, including Fréchet inception distance (FID) [77], Precision, and Recall [78]. As for spectral consistency, we qualitatively evaluate the spectral similarity between the synthetic HSIs and those of the corresponding objects in the original dataset.

5) Conditional Synthesis Experiments Setting: Considering that unconditional synthesis do not have labels, the synthetic HSIs are only applicable to semisupervised or unsupervised tasks without manual labeling. To a certain degree, the unconditional synthesis limits the potential application of the synthetic HSIs. Therefore, we conduct the conditional synthesis experiment to further expand the potential applications of synthesized HSIs. The conditional synthesis experiments include the semantic-guided synthesis and the scene category-guided synthesis. The detailed derivation of conditional synthesis experiments could be found in [36] and [41].

Regarding the semantic-guided experiments, we choose the Xiongan hyperspectral classification dataset [79], which includes 21 object classes, e.g., water, rice, and soybean. For the generation experiments, we chose 60% of the Xiongan dataset for training the UnmixDiff. During the synthetic process, we randomly assign the label value and synthesize the HSIs based on labeling, expanding the training data to three times its original size. For the scene category-guided synthetic experiments, we choose the scene classification dataset HSRS-SC [80]. It contains 1385 HSIs with five scene classes, e.g., agriculture, building, and city building. The HSRS-SC datasets have 256 $\times$ 256 spatial size and 48 spectral bands covering from 380 to 1050 nm. The proportions of the training, validation, and testing are set to 75%, 15%, and 10%, respectively.

B. Ablation Experimental Results

1) Comparison Between the Symmetric AE and the Unmixing Autoencoder in © Synthesis Tasks: In this section, we construct several ablation experiments to evaluate the feasibility and necessity of the UnmixDiff. To demonstrate the necessity of introducing the unmixing autoencoder into the DM, we construct a symmetric autoencoder model to project a hyperspectral image into a low-dimensional latent space and train the DM in this low-dimensional space. The encoder is responsible for learning the latent feature of the HSI, and the decoder part aims to reconstruct the HSI from the latent feature. The DM is constructed based on the latent space learned by the symmetric autoencoder. The comparisons of generated latent space and abundances are shown in Fig. 6. The comparison of the synthetic HSIs by symmetric AE-based DM and UnmixDiff is shown in Fig. 7.

From Figs. 6 and 7, the comparison results show that although symmetric autoencoder models can learn low-dimensional representations of images, enabling the training of DMs, the DMs trained in the latent space cannot effectively capture the characteristics of spatial distribution. Due to the lack of effective constraints on the latent feature generated by the symmetric AE, the feature representation is highly unstable. Moreover, the latent feature generated by the symmetric AE does not have any physical meaning, resulting in the synthetic latent feature not exhibiting meaningful information. Due to the unreliable latent results by

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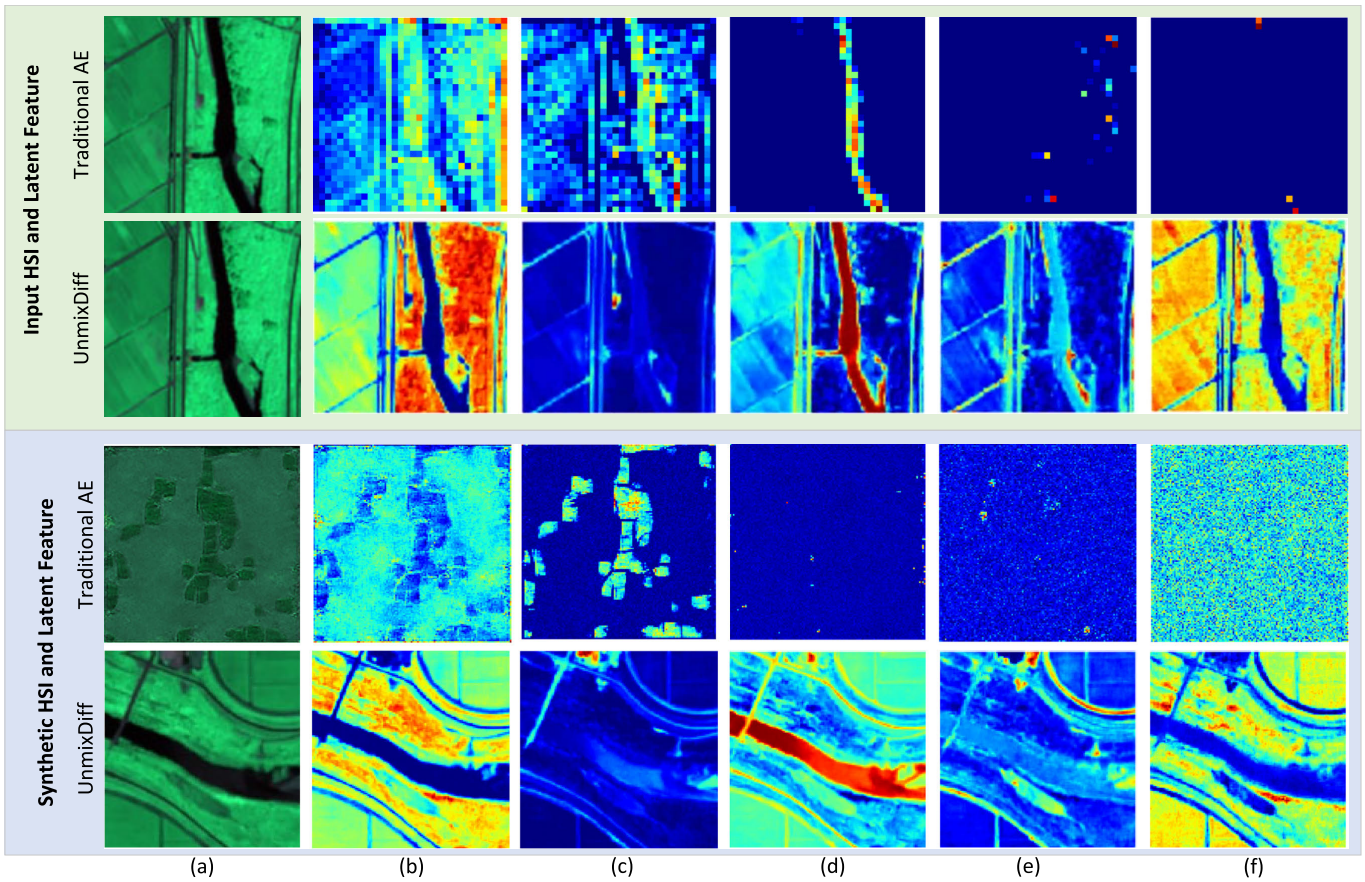


Fig. 6. Comparison of symmetric AE and the proposed unmixing autoencoder. The top half illustrates the latent feature obtained by the symmetric AE and the abundance maps obtained by the unmixing autoencoder. The bottom half illustrates the synthetic latent feature obtained by the symmetric AE-based DM and the synthetic abundance maps obtained by the UnmixDiff. Column (a) represents the RGB synthesis image, and columns (b)–(f) are the abundance maps representing the vegetation, building, water, road, and soil, respectively.

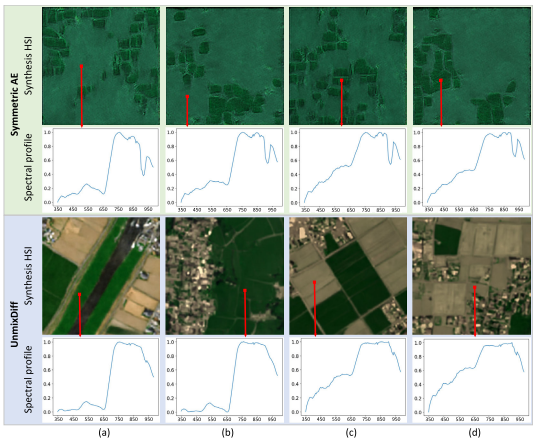


Fig. 7. Comparison of the synthetic HSIs and corresponding spectral profiles by symmetric AE-based DM and the proposed UnmixDiff. Columns (a) and (b) and (c) and (d) represent the spectral profile of synthetic vegetation and soil in the synthetic HSIs, respectively.

the symmetric AE, the corresponding synthetic results cannot recognize useful spatial information. The spatial feature of the synthetic HSIs by symmetric AE-based DM, such as shapes, textures, and colors, do not match the real world. In contrast, the abundance maps reflect the composition and distribution of the materials. The reasonable abundance maps guide the DM to generate physically meaningful and reliable results. Additionally, the synthetic HSIs exhibit strong spectral consistency for the same materials across different HSIs.

It is noteworthy that some traditional dimensionality reduction methods, such as PCA, can be utilized for dimensionality reduction and reconstruction. However, statistical dimensionality reduction methods neglect the physical meaning along with the spectral dimensionality. Moreover, those traditional dimensionality reduction methods usually have a clear range of application, e.g., PCA for linear transformation, which limits the range of applications. In our proposed framework, the unmixing network aims to achieve dimensional reduction and strives to accurately reconstruct the HSI from the low-dimensional abundance information. The proposed UnmixDiff projects hyperspectral images onto the abundance domain, which not only reduces the dimensionality of the HSIs but also assigns physical meanings to the latent generation process. From the ablation experiments, simply learning the low-dimensional features of HSIs through symmetric autoencoders cannot effectively guide the generation process.

2) Parametric Sensitivity Analysis: Fig. 8 shows the unmixing performance under different loss function weights α and β, where (a) and (b) respectively represent the trends of the spatial quality PSNR and the reconstructed SAD with the change in α. We obtain the best PSNR when $\alpha = 0.1$. As for the rSAD, it significantly decreases when the weight of the Loss_{SAD} changes from 0 to 0.1. However, as the α continues to

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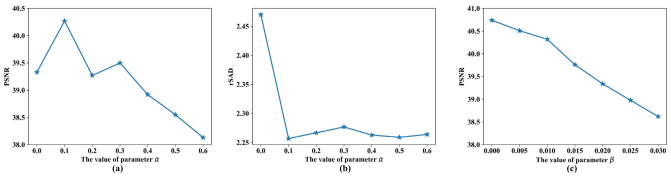


Fig. 8. Unmixing performance under different loss function weights α and β. (a) and (b) represent the PSNR and rSAD evaluation results with different α, respectively. (c) PSNR evaluation results using different β

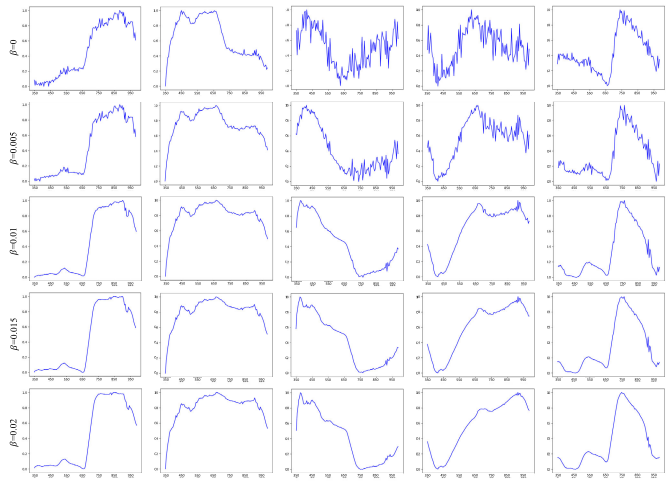


Fig. 9. Extracted end-members under different Loss_{TV} weights.

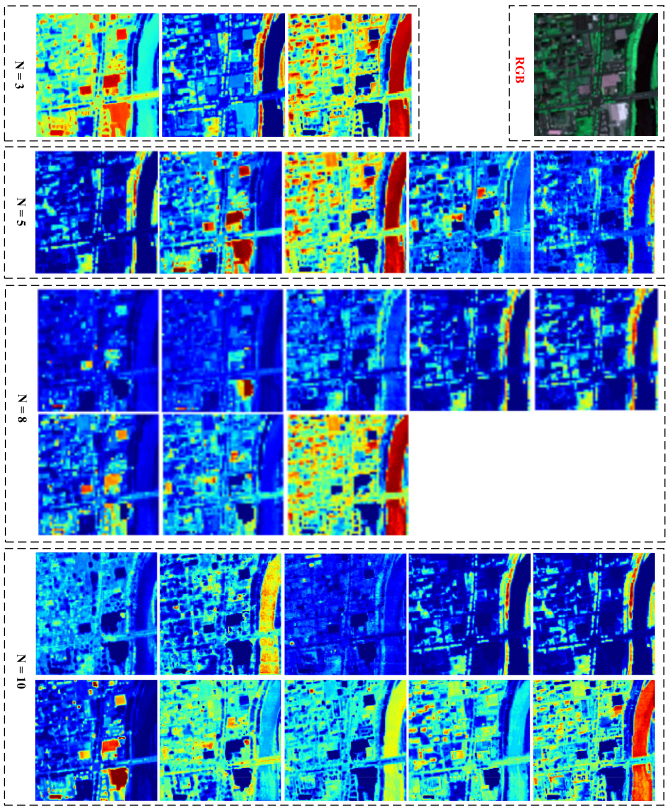


Fig. 10. Abundance maps with different number of end-members.

increase, the rSAD does not significantly improve. Therefore, we chose $\alpha = 0.1$ as the weight of the Loss_{SAD}. Fig. 8(c) represents the trend of the reconstructed image PSNR under different Loss_{TV} weights. It can be seen that the PSNR of the image will decrease as the weight increases. However, discarding the Loss_{TV} would lead to serious distortion of the extracted end-members. Therefore, it is necessary to choose an appropriate parameter to balance the reliability of extracted end-members and the final unmixing performance.

As shown in Fig. 9, when the weight of Loss_{TV} is 0, the unstable random initialization leads to severe distortion of the ultimately extracted end-members. By adding a TV constraint to the end-members, the reliability of the extracted end-members can be ensured, demonstrating the rationality and necessity of the proposed Loss_{TV}. As β gradually increases, the extracted end-members become increasingly smooth, which may lead to the loss of some spectral details. Moreover, as shown in Fig. 8(c), the unmixing reconstruction error gradually increases with the increase in TV weight. Therefore, we chose a weight of 0.01 as our Loss_{TV} weight. We believe that when the Loss_{TV} weight $\beta = 0.01$, it not only ensures the reasonableness of the extracted end-members but also balances the overall unmixing accuracy.

3) Number of End-Members: We evaluate the influence of the number of end-members on the unmixing and synthesis performance. The learned abundance maps under the different number of end-members are shown in Fig. 10. The visualization results of the abundance maps show that the similarity and redundancy between the abundance maps will increase as the dimensionality increases. Therefore, although a higher number of end-members improves the reconstruction accuracy, the abundance maps do not have typical physical meanings, and it also increases the computational complexity of the DM. On the other hand, when the dimension of the abundance maps is low, although it reduces the computational complexity of the abundance DM, there are still mixing phenomenon in abundance maps, resulting in lower quality of the generated results. In the experiments, when the number of end-members is set to 5, the overall framework can guarantee the performance of unmixing reconstruction and the synthesis quality of HSIs simultaneously. Additionally, the visualization results of abundance maps align better with the actual object distributions.

The performance under different numbers of end-members is shown in Table I. Table I shows that the unmixing reconstruction accuracy gradually improves as the abundance maps' dimensionality increases. However, the FID performance of the DM decreases significantly with the increase in the number of end-members. When the dimension of inputs is 31, the synthesis results become pure noise. The reason for the severe decline as the number of end-members increases is the denoising architecture of the DMs. The noise is estimated by the U-Net architecture, resulting in the same dimension of inputs. Therefore, the dimension of inputs directly affects the difficulty of noise estimation. The DM cannot achieve available denoising performance for high-dimensional inputs, which will deteriorate the generation quality. Notably, the normalized synthesis performance decreases when the number of end-members increases from 3 to 5. Since the quality of synthesized HSIs is determined by both the unmixing network and the DM, the inadequate representation of the unmixing network will degrade the overall synthetic perfor-

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TABLE I
SYNTHESIS AND UNMIXING PERFORMANCE UNDER DIFFERENT NUMBERS OF END-MEMBERS

		3	5	8	10	15	31
Unmixing performance	PSNR	35.95	40.32	41.90	42.89	44.48	46.14
	rRMSE	0.020	0.012	0.011	0.010	0.008	0.007
	rSAD	3.04	1.76	1.52	1.30	1.09	1.00
Synthesis performance	FID	3.13	4.41	19.77	36.89	71.70	-
	Precision	0.42	0.51	0.32	0.20	0.11	-
	Recall	0.50	0.52	0.30	0.17	0.08	-
Running Time	Training Time	14h	23h	65h	87h	122h	168h
	Synthesis time	56s	57s	57s	57s	58s	59s

TABLE II
QUANTITATIVE RESULTS OF UNMIXING PERFORMANCE

	Chikusei			Whu-OHS			ARAD		
	PSNR ↑	rRMSE ↓	rSAD ↓	PSNR ↑	rRMSE ↓	rSAD ↓	PSNR ↑	rRMSE ↓	rSAD ↓
DAEU	34.906	0.025	3.566	30.558	0.030	3.317	32.231	0.041	8.211
CNNAEU	35.800	0.020	3.045	31.292	0.026	3.053	33.021	0.037	7.648
MSNet	35.493	0.021	2.944	32.384	0.025	2.706	35.932	0.027	5.754
Ours	40.316	0.012	1.760	37.638	0.016	2.072	39.546	0.023	4.371

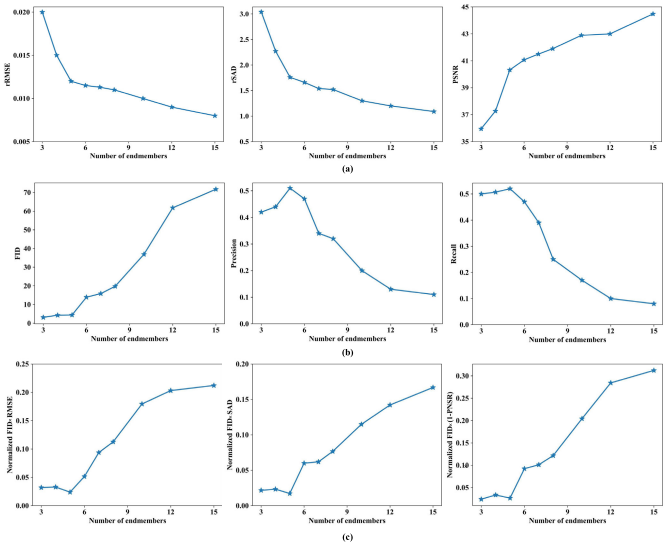


Fig. 11. Different performance metrics changing with the number of end-members. Rows (a)–(c) represent the unmixing, synthesis, and normalized overall performance, respectively.

mance. To better understand the impact of the number of end-members on unmixing and synthesis quality, we plot the line charts showing the performance changing with the number of end-members, as shown in Fig. 11. Moreover, we also obtained a normalized performance of image synthesis quality with respect to the number of end-members. During the normalization process, the min and max values for RMSE, SAD, PSNR, and FID are set to $[0, 0.05]$, $[0, 5]$, $[25, 50]$, and $[0, 50]$, respectively. From the normalized overall performance, when the number of end-members equals 5, we obtain the best-balanced number of end-members that achieve reasonably good unmixing and synthesis performances.

Table I also reflects the running time under a different number of end-members. We can observe that the training time significantly increases with the increase in the number of end-members. When the number of end-members increases from 5 to 8, the training time increases dramatically, costing almost three days. When the number of end-members is 31, even if the training time exceeds a week, the synthetic results are still purely noise. Upon completing the training, the generation time of the abundance map is virtually unaffected by the number of end-members. The generation time for a 256×256 size image requires approximately 1 min.

C. Results

1) Unmixing Performance: Table II shows the quantitative results of unmixing performance on the Chikusei, Whu-OHS, and ARAD datasets, respectively. The visualization of abundance maps and corresponding end-members is shown in Fig. 12.

Because of the complexity of real scenarios, it is impossible to set exactly end-members to represent all the surface materials. The extracted end-member and corresponding abundance maps do not represent a specific material, but rather represent the typical composition of the dataset. Ground object distribution has salient and common low-rank characteristics, empowering a small scale of end-members to describe the entire complex dataset, and the diversity can also be manifested by the numerous abundance maps.

The quantitative evaluation results demonstrate that although traditional unmixing networks have achieved satisfactory unmixing performance on individual hyperspectral images, they exhibit poor generalization performance and fail to produce robust unmixing results in complex scenes. These networks require initialization using end-member initialization algorithms before unmixing. They usually comprise a limited number with simple convolutional layers to meet the single HSI training strategy. In traditional unmixing tasks, due to data limitations, very deep network layers and complex network design may impair the unmixing performance. Therefore, when confronted with hyperspectral datasets comprised of multiple scenes featuring an uncertain number and type of end-members, traditional networks fail to achieve robust unmixing outcomes and satisfactory generalization performance,

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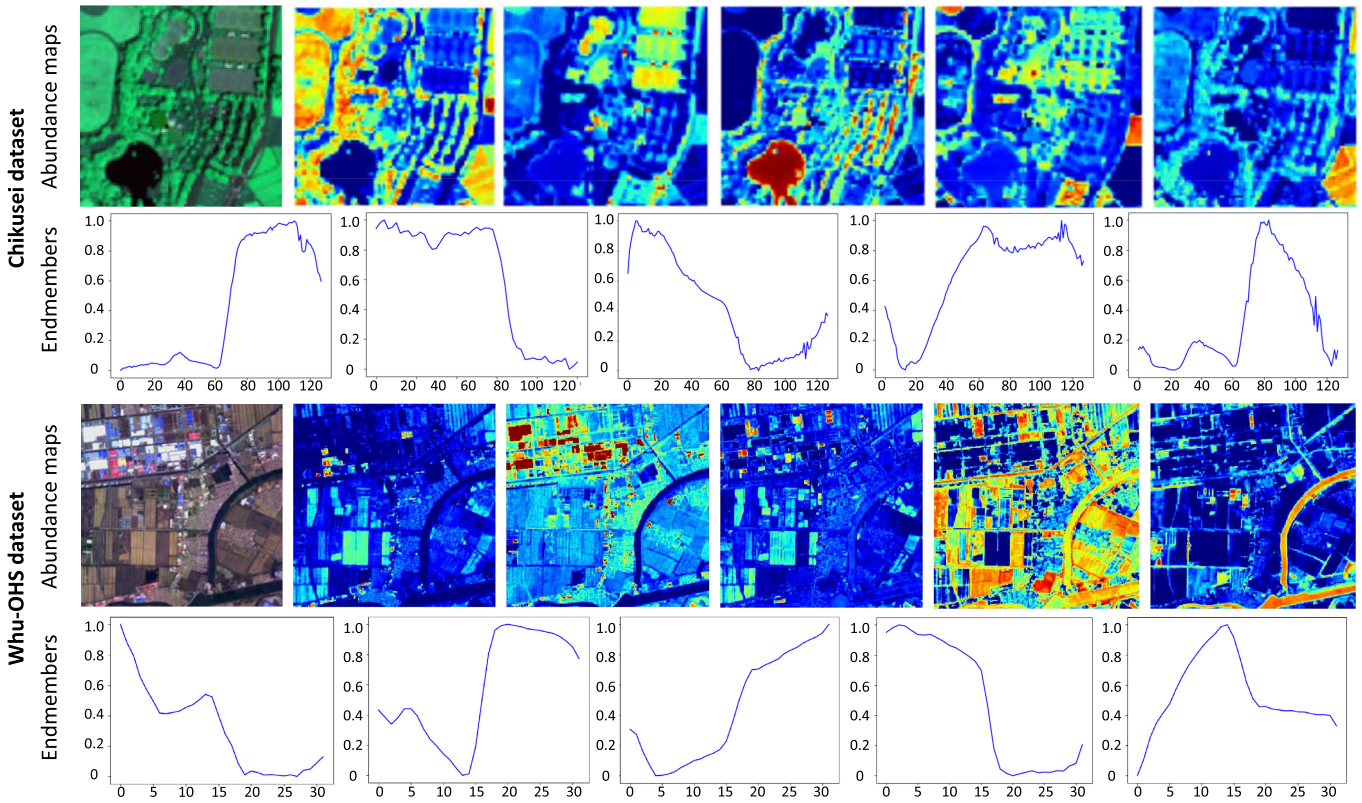


Fig. 12. Visualization of abundance maps and end-members derived by the proposed unmixing framework. The extracted end-members represent the typical composition of the dataset and the abundance maps represent the corresponding proportion.

resulting in significant reconstruction errors. In the domain of hyperspectral image generation, we specifically designed the unmixing network to address the limitations of traditional networks, and the proposed network has demonstrated exceptional unmixing performance. The results obtained from the ARAD dataset demonstrate that in scenes with highly complex land cover types, we can effectively represent the image as a combination of end-members and abundances, leading to accurate reconstruction outcomes. These favorable unmixing results suggest that our unmixing network can acquire low-dimensional image representations from the perspective of abundance, thereby exhibiting strong applicability and generalization performance. Thus, the proposed UnmixDiff model not only suits scenes with a single land cover composition but also performs admirably in scenes featuring complex land cover compositions.

2) Synthesis Performance: Generating hyperspectral images requires maintaining the consistency of their spectral profile, making it a challenging task compared to generating RGB images. The visualization of high-resolution generated HSIs is shown in Fig. 13. Based on the obtained results, it is evident that DM models fail to generate high-dimensional hyperspectral images effectively, resulting in predominantly noisy images without meaningful information. On the other hand, the LDM model addresses this issue by training an autoencoder network to learn the latent features of hyperspectral images, significantly reducing image size and dimensionality. Nonetheless, due to the design characteristics of its autoencoder, the LDM model cannot ensure the spatial and spectral features of the generated images. Consequently, the generated spectral images lack practical significance and do not meet the requirements of hyperspectral image generation. In contrast, our proposed UnmixDiff model can generate high-quality HSIs with realistic spatial details and reasonable object distribution.

TABLE III
QUANTITATIVE RESULTS OF GENERATION PERFORMANCE

	dataset	FID ↓	Precision ↑	Recall ↑
UnmixDiff	Chikusei	4.41	0.51	0.52
UnmixDiff	Whu-OHS	11.77	0.41	0.37
UnmixDiff	ARAD	3.46	0.63	0.46

The visualization comparison of the synthetic spectral profile and the original spectral profile is shown in Fig. 14. Regarding spectral dimensionality, the generated spectral profiles exhibit a high level of consistency with the original spectral profile. The high spectral consistency of the same objects in different HSIs also demonstrates the effectiveness of the proposed method in mitigating the spectral variability problem. The overall experiments demonstrate the reliability of our proposed method in hyperspectral image generation.

The quantitative results of generation performance are shown in Table III. Since the DM and LDM cannot generate meaningful results, we argue that it is meaningless to evaluate the quality of image generation, and we only evaluate the effect of the proposed method on different datasets. From the quantitative results, it can be observed that the proposed methods can generate high-quality HSIs with rich diversity on different datasets.

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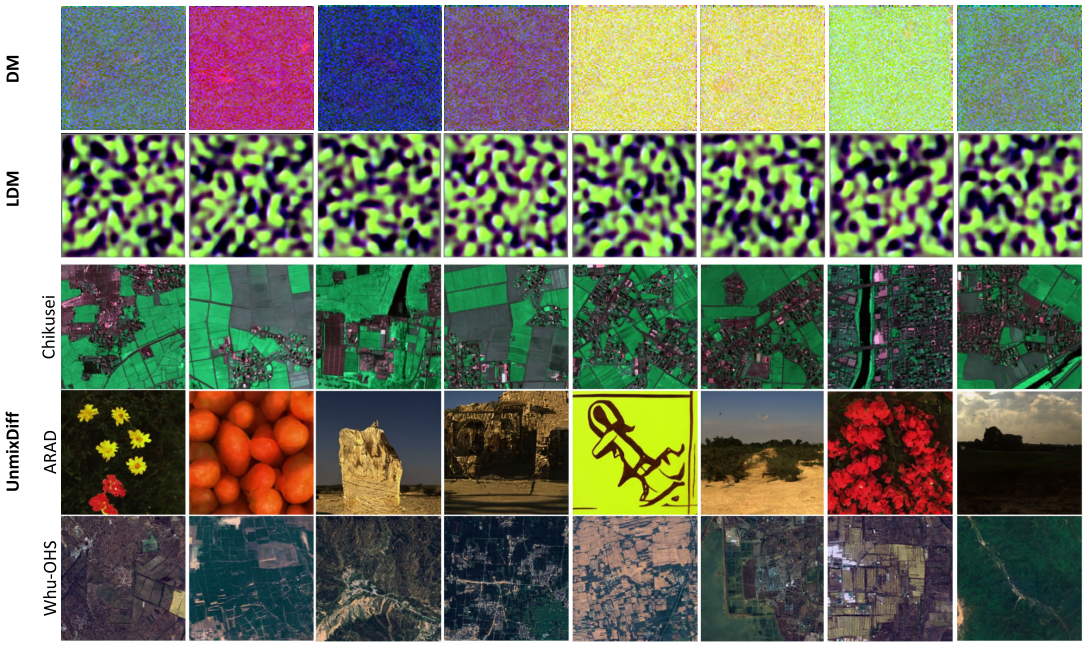


Fig. 13. Comparison of synthesis results. The DM and LDM cannot generate meaningful synthesis results. In contrast, our method shows desirable performance.

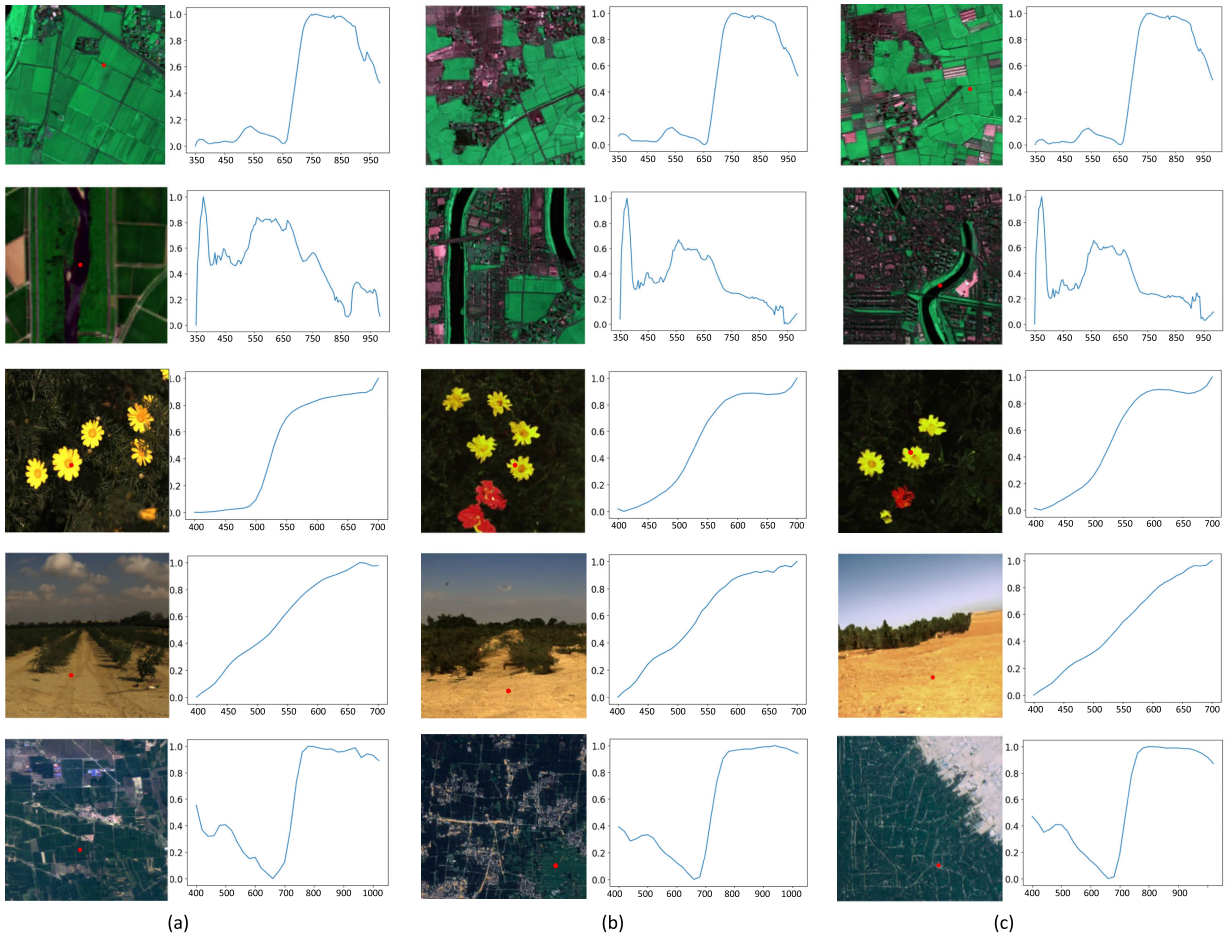


Fig. 14. Visualization comparison of the synthetic spectral profile and the original spectral profile. The x-axis and y-axis represent the wavelengths and reflectance, respectively. Column (a) represents the existing HSI and corresponding spectral profile. Columns (b) and (c) represent the synthetic HSIs and corresponding spectral profiles. The red dot represents the sampling position of the spectral profile.

Furthermore, we also compare synthetic performance between the proposed UnmixDiff and the state-of-the-art HSI synthetic method MSF-Diff [58], as shown in Fig. 15. From Fig. 15, the MSF-Diff only supports the visible wavelength

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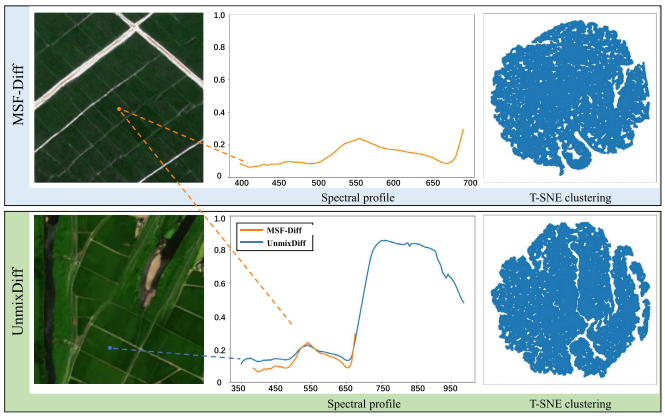


Fig. 15. Synthetic comparison between the MSF-Diff and the proposed UnmixDiff.

range, neglecting the spectral feature beyond the visible bands. The MSF-Diff only synthesizes a narrow spectral range, constituting a small part of the proposed UnmixDiff. Subsequently, we conducted T-SNE clustering to compare the divisibility of the MSF-Diff and UnmixDiff. The narrow spectral bands reduce the divisibility of the synthetic HSIs. The MSF-Diff reveals relatively poor divisibility due to the absence of the spectral information in other bands. Only synthesizing the 400–700 nm HSIs greatly limits the downstreaming applications. In contrast, the proposed UnmixDiff can synthesize high-quality HSIs covering the entire spectral range and meet the demands for wide-band HSI synthesis.

3) Conditional Synthesis Experiments: In this section, we conduct the conditional HSI synthesis experiments to control the synthetic contents. Specifically, we conduct the semantic-guided and the scene category-guided conditional synthesis experiments.

The visualization of the semantic-guided synthetic results is shown in Fig. 16. By introducing the semantic labels into sampling process, we can easily control the content of the synthetic results. Moreover, we can modify the labels to further control the synthetic contents. From Fig. 17, the synthetic content changes along with the adding or removing of the label regions. The semantic-guided synthetic results demonstrate the potential of the UnmixDiff in the conditional synthetic tasks.

After that, we conduct the classification experiments based on the original dataset and the corresponding synthetic HSIs. For the original dataset, the previous selected 60% data is used for training the classification model, and the rest of 20% for validation, and 20% for testing. Consequently, we conduct the classification based on the original dataset and the synthetic HSIs. To verify the reliability of the synthetic data, we choose three typical hyperspectral image classification methods, including the SVM, 1DCNN, and 2DCNN. To evaluate the classification performance, we choose the commonly used metrics, including accuracy, $F1$-score, and Kappa to evaluate the classification results. The quantitative results are shown in Table IV.

From Table IV, the proposed method can effectively improve the data scale and diversity by synthesizing high-quality HSIs, which can facilitate the model learn more robust features. In the classification tasks, the sample size

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS WITH/WITHOUT SYNTHETIC HSIS

Dataset	Data scale	Metrics	SVM	1DCNN	2DCNN
Xiongan	2.2M	Accuracy	71.173	83.466	97.846
		F1-Score	0.664	0.783	99.280
		Kappa	0.709	0.818	0.975
Xiongan+Syn	6.6M	Accuracy	75.688	86.393	97.863
		F1-Score	0.697	0.811	99.285
		Kappa	0.742	0.843	0.976

TABLE V
OVERALL SCENE CLASSIFICATION ACCURACY OF DIFFERENT MODELS WITH/WITHOUT SYNTHETIC HSIS

Dataset	AlexNet	VGGNet	ResNet
HSRS-SC	87.43%	65.64%	68.92%
HSRS-SC+Synthetic	89.77%	66.50%	76.07%

for different objects often vary greatly. By editing the label category, the method can effectively augment the sample size of some few-shot objects. The increase in sample quantity and diversity has a significant promotion effect on both traditional and DL-based methods. The overall performance demonstrates the effectiveness of the proposed framework in the classification tasks.

Besides the semantic-guided synthetic experiments, we also conduct the scene category-guided synthetic experiments and demonstrate the beneficial impact of synthetic data on downstream tasks, particularly scene classification. We use the HSRS-SC [80] datasets to conduct the scene category-guided synthesis experiments and conduct the following scene classification experiments. In the HSRS-SC dataset, different scene categories contain different numbers of HSIs, ranging from 154 to 485. Due to the lower image quality of the unclassified and water, this section only selects agriculture, building, and city building to construct scene classification experiments. The scene category-guided synthetic results are shown in Fig. 18. During the downstream scene classification experiments, we augmented the data with synthetic HSIs until each category contained 1000 samples. After that, we conduct the scene classification experiments using three typical models on this expanded dataset. The overall classification results are shown in Table V. From Table V, the AlexNet achieves the best scene classification results on the original and the expanded dataset. Moreover, the expanded dataset achieves better scene classification performance on the three selected model, demonstrating the effectiveness of the synthetic HSIs. The improvement can mainly be attributed to the expansion of sample size and the mitigation of imbalance issues. The overall performance suggests that synthetic data can enhance the diversity and scale of existing datasets, mitigate issues like sample scarcity and class imbalance, and potentially benefit other downstream tasks.

V. DISCUSSION

The proposed UnmixDiff achieves good performance in the task of hyperspectral image synthesis. The positive results

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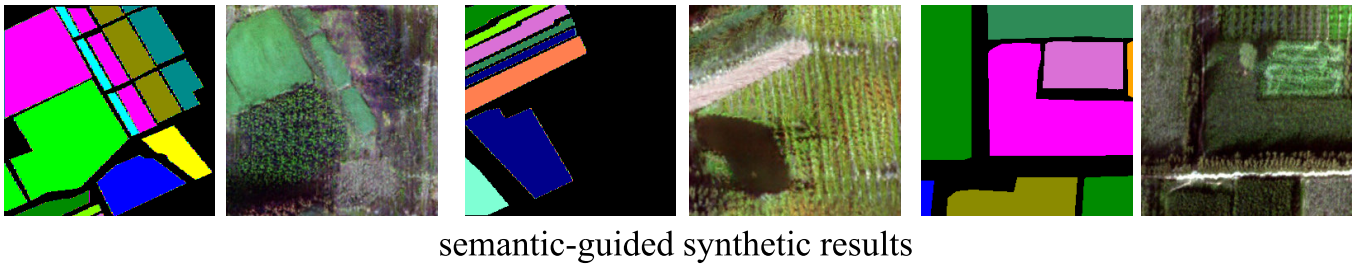


Fig. 16. Semantic-guided synthetic results. The label is obtained from the Salinas Valley and the Xiongan hyperspectral classification dataset.

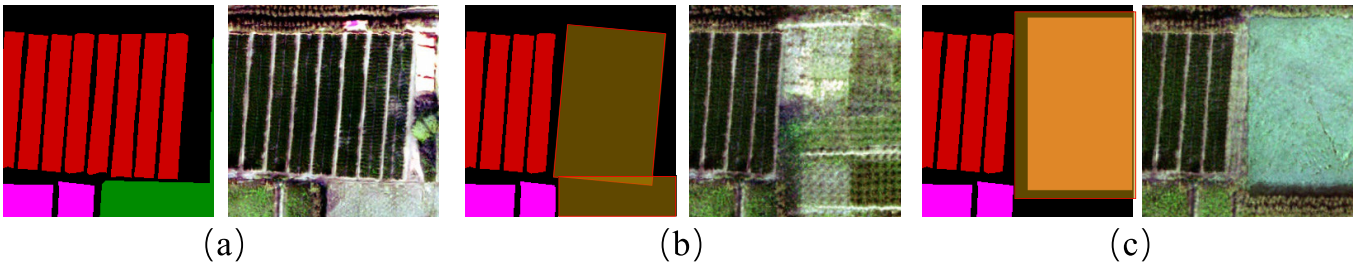


Fig. 17. Editing the semantic labels to control the synthetic contents. (a) Original label and the synthetic results. (b) and (c) Synthetic results after removing and adding some regions.

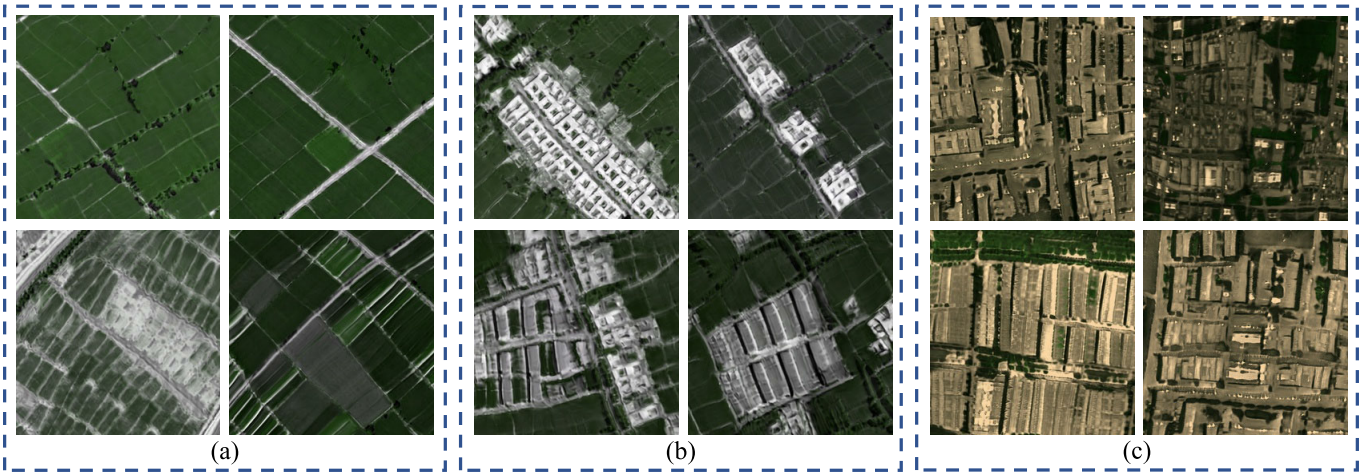


Fig. 18. Scene category-guided synthetic results. (a) Agriculture. (b) Building. (c) City building.

stem from two critical points: the design of the unmixing network for the synthesis tasks and the conducting of the abundance-based DM.

Regarding the unmixing network, our method has achieved excellent results compared to the traditional single HSI unmixing network. We attribute the improvement to the redefinition of the unmixing task from the perspective of hyperspectral synthesis tasks. From the perspective of task definition, the proposed UnmixDiff integrates the unmixing process as an element of the HSI generation task rather than an individual unsupervised process. The redefinition of the unmixing process guides the network designing process. From the training perspective, our unmixing model can learn general network parameters across the dataset, enhancing network generalization and robustness. The proposed framework abandons the characteristics of traditional unmixing methods that train and adjust parameters on a single HSI.

In the subsequent synthesis tasks, we construct the DM in the low-dimensional abundance domain, solving the problem that traditional DMs cannot directly synthesize hyperspectral images. Conducting the abundance-based DM reduces the computational complexity, increases the precision of noise inferences, and introduces clear physical significance to the hyperspectral synthesis tasks. The abundance-based DM uses the U-Net architecture to recover the abundance maps from the random noise. The characteristics of U-Net architecture determine that the input and output should have the same dimensions. An excessively high number of end-members will severely influence the performance of the DM. Therefore, considering the impact of unmixing accuracy on the final generation quality, we have constructed many ablation experiments and obtained the optimal number of end-members. From the synthesis results, the synthetic HSIs have very realistic spatial details and good spectral consistency of the same materials in different HSIs.

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However, this article mainly focuses on synthesizing HSIs with realistic land cover distributions and consistent spectral characteristics with the original dataset, neglecting the reasonableness of the original spectral profile. The HSIs are often affected by noise, water vapor absorption, atmospheric effects, and other factors. Therefore, in practical applications, we need to use preprocessing methods such as polishing, denoising, and band selection operation to ensure the rationality of the original spectral profile. In addition, the spatial–spectral resolution of the synthetic results highly depends on the input data size and the parameter settings. Therefore, the proposed method is hard to generate very large size, e.g., $10\,000 \times 10\,000$ HSIs, which weaken the flexibility and the generic of the framework.

VI. Conclusion

This article proposes UnmixDiff to solve the feasibility of deep generative models in HSI synthesis tasks. The method synthesizes the HSIs from the perspective of material distribution, introducing a clear physical interpretation into the hyperspectral image synthesis tasks. By introducing an unmixing autoencoder into the diffusion generation model, the UnmixDiff projects the hyperspectral image synthesis from the high-dimensional spectral domain to the low-dimensional abundance domain, which greatly reduces the computational complexity and effectively improves the spectral confidence of generated hyperspectral images. By constructing the diffusion generation model in the abundance space, the proposed method focuses on synthesizing the intrinsic distribution of materials rather than superficial textures. Based on this, the UnmixDiff can generate HSIs with realistic spatial details while maintaining the spectral consistency. The proposed UnmixDiff achieves excellent performance in hyperspectral image synthesis tasks. Moreover, the proposed UnmixDiff can be a fundamental framework for DM-based hyperspectral image processing tasks.

Although the proposed UnmixDiff has shown good performance in hyperspectral synthesis tasks, there are still areas for improvement. From the perspective of model building, the pretrained model requires multiple inference steps to transform random noise into abundance maps. Generating abundance maps with the size of 256×256 requires approximately one minute. Reducing the inference steps and accelerating the generation speed are worthy issues to consider. Recently, the multimodality model has achieved impressive synthetic results. Therefore, adding multimodality tags, e.g., text prompt, will be a promising work to synthesize desired HSIs.

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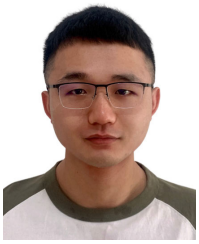
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